Programming for Finance

Introduction to SQL

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Signs

– Please take pictures of the information. Medium priority.

Source: https://www.flaticon.com/ru/free–icon/camera–flash_3460307

– Please write down the information. High priority.

Source: https://openmoji.org/library/emoji–270D/

TOPICS

1. What is SQL
2. SQL Standards and Dialects
3. Database Types
4. SQL Command Types
5. Database Management Tools
6. Python Libraries for SQL

What is SQL

- SQL stands for ***Structured Query Language***
- A standard language for accessing and manipulating databases
  - allows users to communicate with a **database**
  - **declarative**: you tell the database what you want, not how to get it
  - used **to retrieve, update, insert** and **delete** data

https://ru.wikipedia.org/wiki/SQL#/media/%D0%A4%D0%B0%D0%B9%D0%BB:Sql_data_base_with_logo.svg

SQL Standards and History

- Developed in the 1970s by **IBM**
- By 1986, ANSI (American National Standards Institute) and ISO (International Organization for Standardization) standard groups officially adopted the standard "Database Language SQL" language definition
- Most databases comply with the ANSI SQL standard
- New versions of the standard were published in 1989, 1992, 1996, 1999, 2003, 2006, 2008, 2011, 2016 and, most recently, 2023

ANSI
American National Standards Institute

https://en.wikipedia.org/wiki/American_National_Standards_Institute

SQL Standards and History

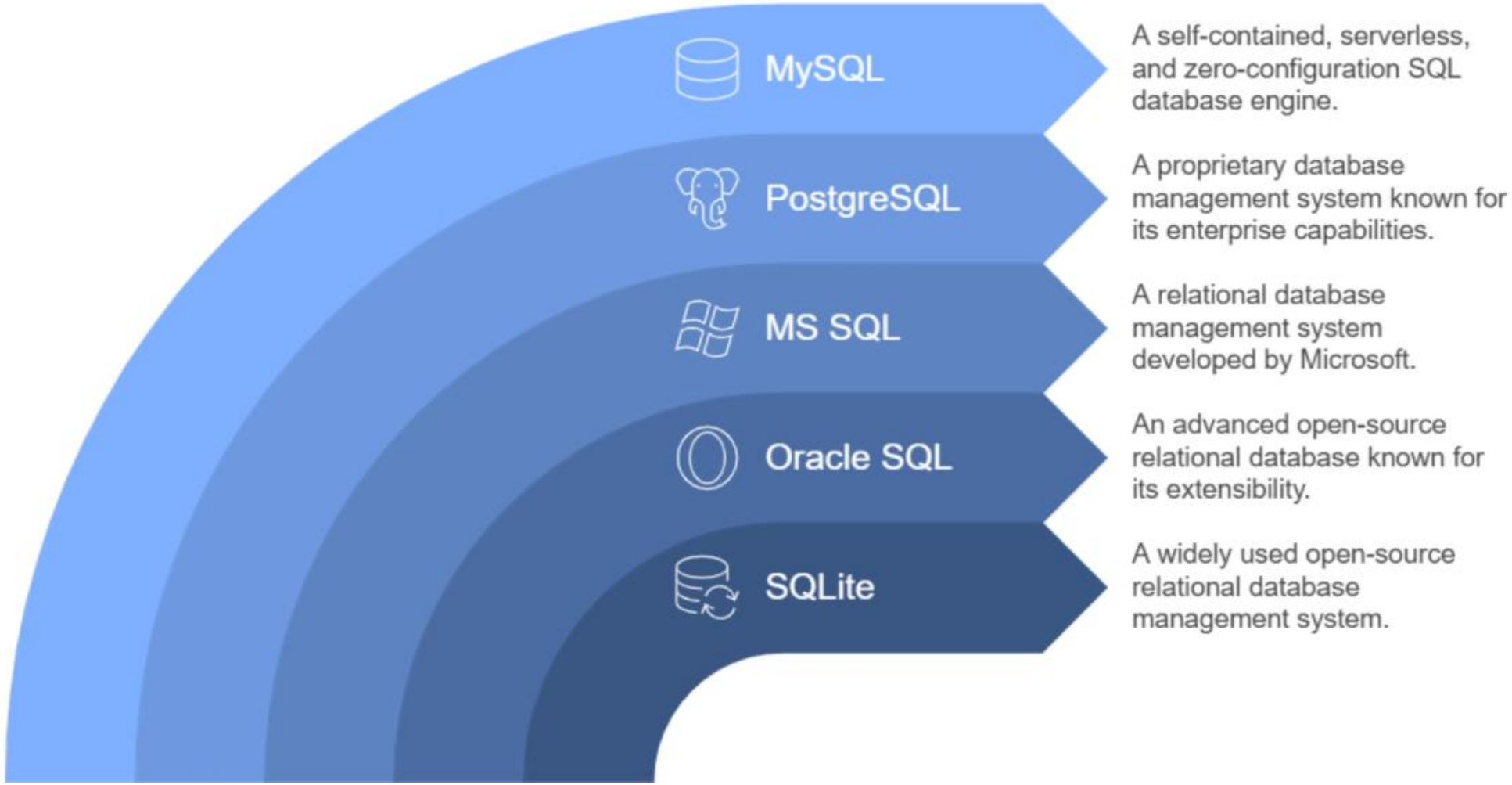
- There is only **one** SQL language standard
- Variations exist called **dialects**
- Core syntax remains similar across systems
- Understanding standard SQL allows to work with most databases

https://www.zeluslugi.ru/info-czentr/it-glossary/term-sql

Common SQL Dialects

- **MySQL:** popular for web applications
- **PostgreSQL:** very popular for managing both enterprise and individuals' databases
- Microsoft SQL Server (T-SQL)
- Oracle Database (PL-SQL)
- SQLite

Which SQL Dialect to Learn?



https://www.linkedin.com/pulse/what-sql-dialect-which-one-should-you-learn-learnsql-com-coelc

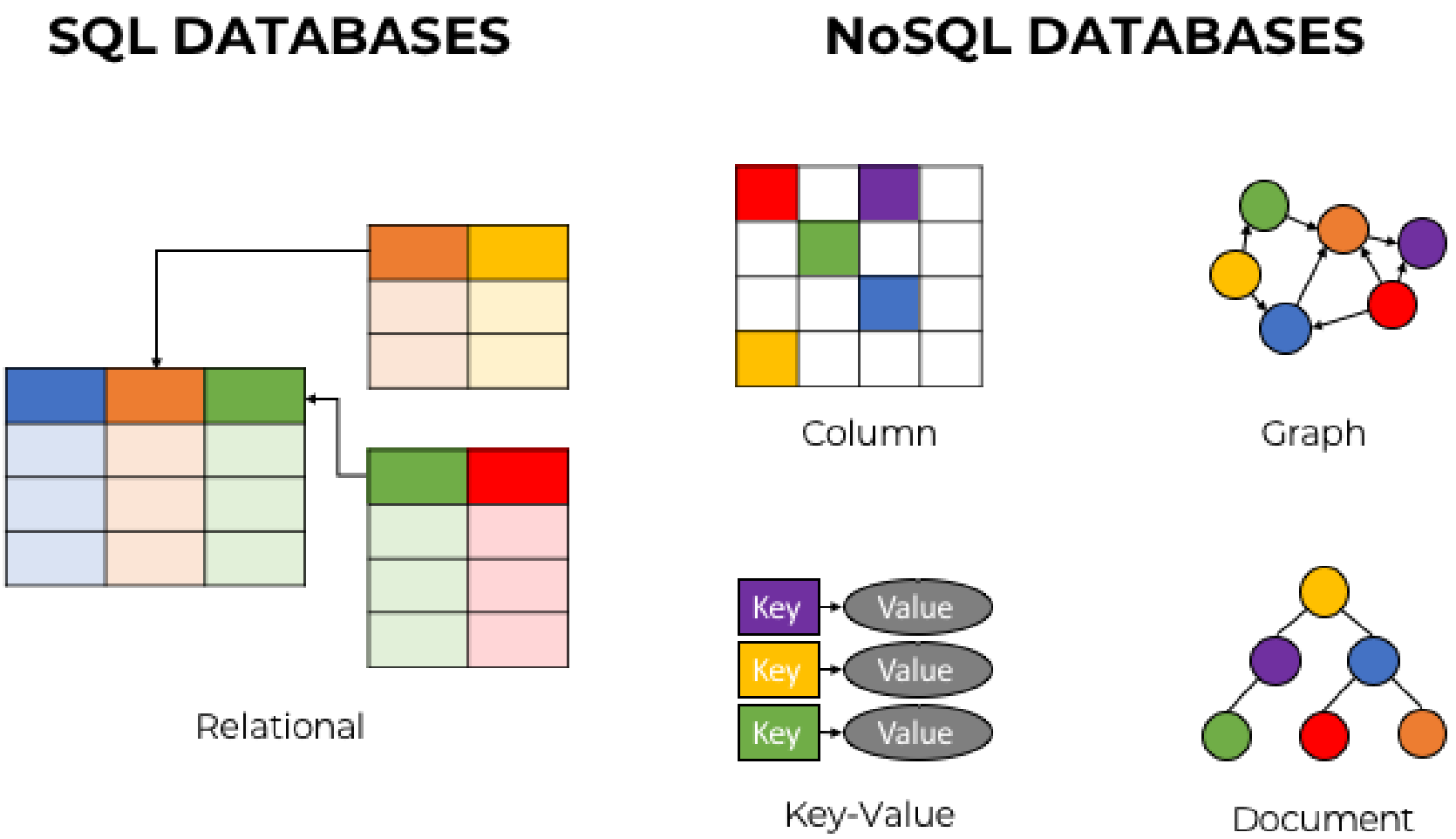
What is SQL Used For

- Querying data for reports and analysis
- Updating records in a database
- Inserting new data entries
- Deleting obsolete or incorrect data
- Creating and modifying database structures
- Managing user access and permissions

https://www.geeksforgeeks.org/sql/introduction-to-sql/

Types of Databases

- Databases store organized data
- Two main types:
  - Relational
  - Non-Relational
- **Relational databases** store data in tables with rows and columns
- **Non-Relational databases** store data in documents, graphs, or key-values
- Choice depends on data structure and scalability needs



Comparison of the relational SQL database and NoSQL databases

https://openclassrooms.com/en/courses/5671741-design-the-logical-model-of-your-relational-database/6255746-compare-relational-and-nosql-databases

Relational Databases (RDBMS)

- Based on the relational model proposed by **E.F. Codd**
- Data is organized into tables with **defined relationships**
- Uses **Primary Keys** and **Foreign Keys** to link tables
- Ensures data integrity and consistency
- Examples: PostgreSQL, MySQL, Oracle, SQL Server

*A **Primary Key** is a special constraint (column or group of columns) in SQL that uniquely identifies each record (row) in a database table.*

*A **Foreign Key** is a column (or group of columns) in one table that points to a **Primary Key** (or a unique key) in another table.*

https://pngtree.com/so/golden-key

Non-Relational Databases (NoSQL)

- Designed for **unstructured** or **semi-structured** data
- High flexibility of data storage
- Does not typically use fixed table schemas
- Usually do not use standard SQL for queries

Why SQL is Useful for Financial Analysis

- For finance, Relational Databases are the standard
- Data integrity and transactional safety are critical
- SQL handles large datasets beyond Excel limits
- SQL ensures data integrity and audit trails
- SQL allows reproducible analysis scripts
- SQL efficiently joins data from multiple sources
- SQL is a standard skill in banking, fintech, and accounting

https://www.freepik.com/vectors/green-tick

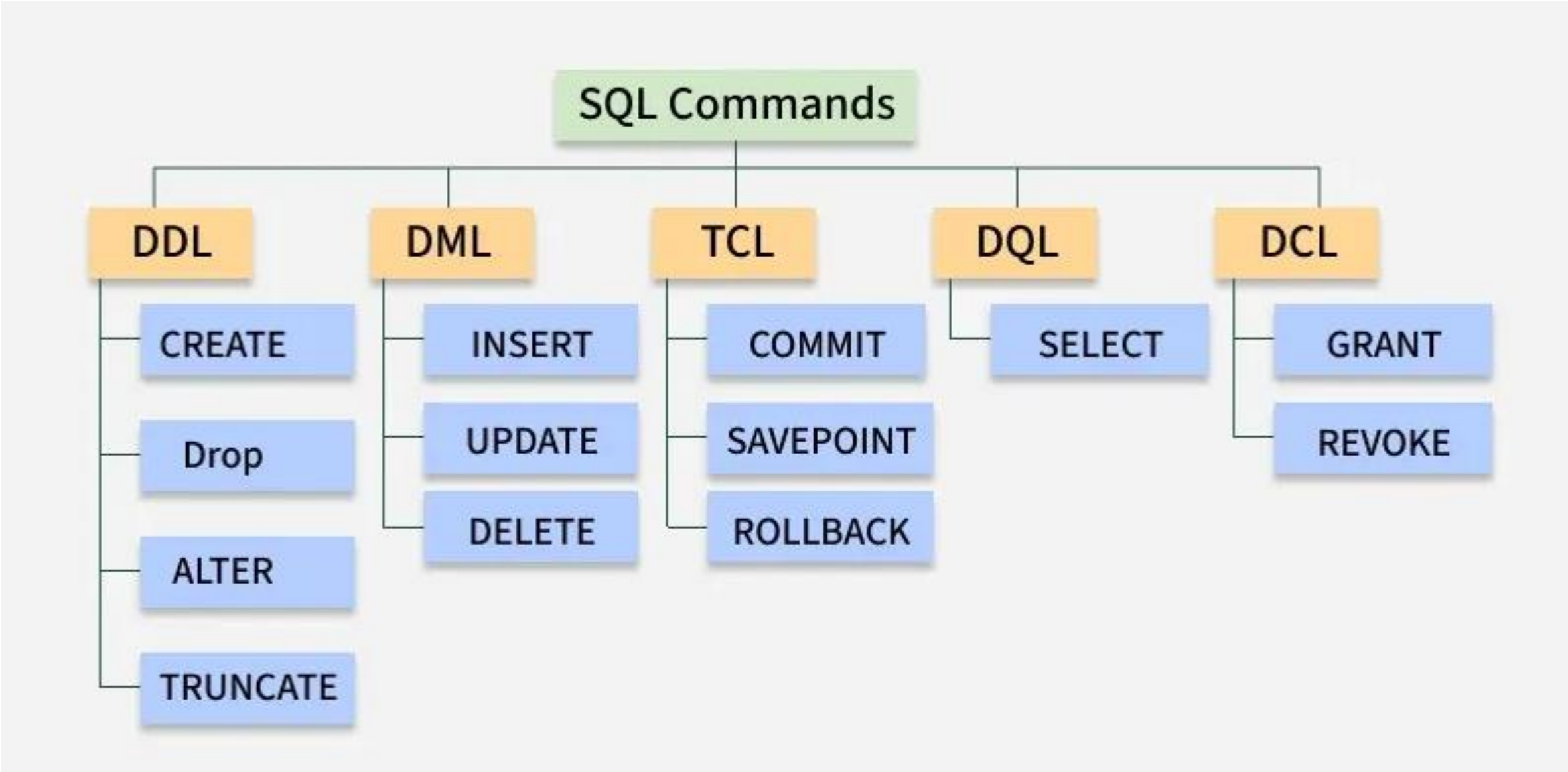
SQL vs Excel in Finance

- **Excel:** Good for ad-hoc analysis, small data, visualization ✅
- **SQL:** Good for large data, automation, data extraction ✅
- **Excel** can crash with millions of rows ❌
- **SQL** databases handle billions of rows efficiently ✅
- *Best practice:* use SQL to extract and aggregate, Excel to visualize

https://es.pinterest.com/pin/115200677968621378 2/

SQL Command Types Overview

- SQL commands are **grouped by function**
- Main groups: **DDL**, **DML**, **DQL**, **DCL**, **TCL**
- Analysts mostly use **DQL** and some **DML**
- Admins and Developers use **DDL** and **DCL**



https://media.geeksforgeeks.org/wp-content/uploads/20250823151715831016/sql_commands.webp

DDL: Data Definition Language

- Defines the database structure
- Commands: *CREATE, ALTER, DROP, TRUNCATE*
  - Example: CREATE TABLE clients (id INT, name VARCHAR)
- Used by database architects and developers
- Rarely used by financial analysts during daily work

DML: Data Manipulation Language

- Manages data within the objects
- Commands: *INSERT, UPDATE, DELETE*
  - Example: INSERT INTO clients VALUES (1, John)
- Changes the actual data stored
- Requires caution to avoid data loss

DQL: Data Query Language

- Retrieves data from the database
- Command: *SELECT*
  - Example: SELECT * FROM clients
- Most important command for financial analysts
- Used to generate reports and extract datasets

DCL: Data Control Language

- Controls access to data within the database
- Commands: *GRANT, REVOKE*
  - Example: GRANT SELECT ON table TO user
- Managed by Database Administrators
- Ensures security and compliance

TCL: Transaction Control Language

- Manages transactions in the database
- Commands: *COMMIT, ROLLBACK, SAVEPOINT*
- Ensures data integrity during operations
- Critical for financial transactions (e.g., money transfers)
- Ensures all steps succeed or none do

Database Management Tools: Windows

- Microsoft SQL Server Management Studio (SSMS)
- MySQL Workbench
- DBeaver (Community Edition)
- Navicat for SQL Server or MySQL
- pgAdmin 4 (PostgreSQL)

https://en.wikipedia.org/wiki/Microsoft_Windows#/media/File:Windows_logo_and_wordmark_-_2021.svg

Database Management Tools: Mac

- MySQL Workbench (Limited support on newer macOS)
- DBeaver (Community Edition)
- TablePlus
- Sequel Ace (for MySQL)
- pgAdmin 4 (PostgreSQL)

macOS

https://en.wikipedia.org/wiki/MacOS#/media/File:MacOS_wordmark_(2017).svg

Cross-Platform Tools

- DBeaver
- DataGrip
- VS Code
- DbVisualizer
- pgAdmin 4

DbVisualizer

Integrating SQL with Python

- Python is popular for financial **modeling and automation**
- SQL handles data **storage and retrieval**
- Python handles complex logic and machine learning
- Combination creates powerful analytical pipelines
- *Data is fetched via SQL, processed via Python*

Pandas and SQL Integration

- *Pandas* library has read_sql and to_sql functions
  - Allows loading SQL query results directly into DataFrames
  - Allows writing DataFrames back to database tables
  - Simplifies data manipulation after extraction
- Other Python Libraries for SQL: *sqlite3, psycopg2, pymysql, cx_Oracle, pyodbc, SQLAlchemy*

Summary

- SQL is the standard for relational database communication
- Finance relies on Relational Databases for data integrity
- Analysts focus on DQL and some DML
- Python integration extends SQL capabilities for finance

SQLZoo

- SQLZoo includes tutorials and reference to support people learning SQL.
- URL: https://sqlzoo.net/wiki/SQL_Tutorial